Rotaract Club of Xiamen University Malaysia | Rotary Club of Kuala Lumpur Dynamics | Rotary Club of Tun Razak Exchange

SDC Hackathon

Technical Tools Declaration

Hosted by Xiamen University Malaysia (XMUM) AI Club
Submission Deadline: 25 October 2026

Instructions: Declare every API, cloud service, AI tool, dataset, model, and sponsor resource your team used to build the project. Complete every section; where a category does not apply, tick "No" or write "None" – do not leave it blank. Submit as a single PDF document named "SDC2026_TeamName_TechDeclaration.pdf" together with your project submission. Undeclared third-party resources discovered at judging may affect your score or eligibility.

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SDC HACKATHON 2026

Technical & Tools Declaration

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1. Team & Project Overview

Team Name	Small x Big y
Project / Solution Name	RestockAI
Name (Leader)	Eishvar A/L Rajesh Khanna
Email & Phone (Leader)	eishvar2020@gmail.com, 0175455258
University Affiliation(s)	AIMST University
Live Prototype URL	https://restockiq.replit.app
Code Repository URL (public / access granted to judges)	https://github.com/Eishvar/restockiq

2. Architecture & Technology Stack

2.1 System Architecture Overview

Describe how the components declared in this document connect end-to-end (e.g., user interface → backend → AI model/API → database → cloud hosting). Attach a system block diagram as an appendix if available.

[Maximum 200 words]

Users open the React web app in a browser and choose a demo role (owner, manager or staff). The app calls a FastAPI backend over HTTPS. The backend reads and writes a PostgreSQL database (sales, stock batches, recipes, suppliers, plans, orders, activity log). When someone presses Generate, the backend runs a planning engine written with pandas and NumPy. It forecasts demand per dish, converts it into ingredient needs, subtracts unexpired stock, adds a safety buffer and rounds up to supplier packs. This engine contains no AI. A manager submits the order, the owner approves it, and every step is logged. A language model is reached through the OpenRouter API in two places only: the Help chat, and reading uploaded PDF sales reports. In the PDF case OpenRouter's PDF reader extracts the text, the model copies the sales lines, and a person checks them before saving. Everything runs as one Python process on a Replit Autoscale deployment with Replit's PostgreSQL

2.2 Languages, Frameworks & Libraries

List the main programming languages, frameworks, and third-party libraries that your team did not create. Open-source use is permitted provided it is declared and properly licensed/attributed.

Name & Version	Purpose in Project	License
Python 3.12	Backend and planning engine	PSF License
TypeScript 6.0 (run on Node.js 20)	Frontend language and build tooling	Apache-2.0 (TypeScript), MIT (Node.js)
FastAPI 0.159	Web API	MIT
Uvicorn 0.54	Web server	BSD-3-Clause
SQLAlchemy 2.1	Database access	MIT
Pydantic 2.14	Data checks, strict AI answer format	MIT

Name & Version	Purpose in Project	License
pandas 3.0, NumPy 2.5	Forecast and purchasing maths	BSD-3-Clause
psycopg2-binary 2.9	PostgreSQL driver	LGPL with exceptions
python-dotenv 1.2	Local settings	BSD-3-Clause
openpyxl 3.1	Reading Excel sales files	MIT
python-multipart 0.0.32	File uploads	Apache-2.0
OpenAI Python SDK 3.36	Calling OpenRouter	Apache-2.0
Anthropic Python SDK 2.1	Optional direct Claude call (coded, only used if configured)	Apache-2.0
tzdata 2026.5	Time zone data	Apache-2.0
pytest 9.1, httpx 0.28	Automated tests	MIT, BSD-3-Clause
matplotlib	Draws one chart in the data script only	Matplotlib licence (PSF-style)
React and React DOM 19.3	User interface	MIT
React Router 7.18	Page navigation	MIT
TanStack Query 5.104	Loading and caching data	MIT
Recharts 3.10	Charts	MIT
Vite 8.3	Frontend build	MIT
Tailwind CSS 4.3	Styling	MIT

Name & Version	Purpose in Project	License
oxlint 1.89	Code checker	MIT
DM Sans and Manrope fonts	Text fonts, loaded from Google Fonts	SIL Open Font License

3. APIs & Cloud Services

3.1 External APIs

List every external API your solution calls (GenAI, maps, payments, messaging, data feeds, etc.). Do not include API keys or secrets in this document.

API & Provider	Purpose in Project	Access Tier (free / paid / sponsor-provided)
OpenRouter chat API (OpenRouter)	Help chat answers for questions that are not on the stored list	Paid
OpenRouter PDF reader, OCR engine (OpenRouter)	Reads uploaded PDF sales reports so the AI can copy the sales lines	Paid
Google Fonts (Google)	Loads two fonts in the browser	Free
WhatsApp click-to-chat link, wa.me (Meta)	Opens a pre-written supplier message in the person's own WhatsApp. The app sends nothing itself.	Free

3.2 Cloud Platforms & Deployment

List every cloud service used for hosting, storage, databases, authentication, or compute, and indicate how the prototype is deployed.

Service & Provider	Purpose in Project	Plan / Region
Replit Autoscale deployment (Replit)	Hosts the live app (web pages and API in one process)	Replit Core plan
Replit PostgreSQL (Replit)	Database for the deployed app	Replit Core plan
Replit Secrets (Replit)	Keeps the AI key out of the code	Replit Core plan
GitHub (Microsoft)	Code repository	Free

4. AI Tools, Models & Datasets

4.1 AI Models Used in the Solution

List every AI model embedded in or called by your application, including pre-trained, fine-tuned, and self-trained models.

Model & Version	Source / Provider	Usage (pre-trained / fine-tuned / self-trained)	License / Terms
deepseek/deepseek-v4.1-flash	OpenRouter and the model's own provider	Pre-trained, used as is (Help chat and PDF sales-report reading)	MIT licence (open weights, DeepSeek); accessed through OpenRouter under OpenRouter's terms of service
Mistral OCR (OpenRouter's PDF text-reading engine)	OpenRouter / Mistral AI	Pre-trained, used as is (text from PDFs)	OpenRouter and provider terms
None fine-tuned or self-trained. The forecast is plain statistics, with no trained model.	Team	n/a	n/a

4.2 Datasets

List all datasets used for training, fine-tuning, evaluation, or in the running application. State clearly if a dataset contains personal or sensitive data.

Dataset Name	Source / URL	Type (public / private / self-collected / synthetic)	License	Personal Data? (Y/N)
Demo sales, menu, recipes, suppliers, stock and calendar	Generated by the team's scripts in data/ (github.com/Eishvar/restockiq)	Synthetic	Apache-2.0 (repository licence)	N

Dataset Name	Source / URL	Type (public / private / self-collected / synthetic)	License	Personal Data? (Y/N)
Sample sales report for 22 Oct 2026	Generated by the team's script (data/demo_reports/)	Synthetic	Apache-2.0 (repository licence)	N
Malaysian public holidays (calendar_my.csv)	Compiled by the team from public holiday dates	Public	Public information	N
Simulated restaurants for the forecast audit	Generated by the team (research/forecast_audit/)	Synthetic	Apache-2.0 (repository licence)	N

4.3 AI-Assisted Development Tools

Tools used to help write code, design, content, or documentation (e.g., chat assistants, code completion tools, image generators). Declaring these tools does not reduce your score; undeclared use may.

Did your team use any generative-AI tool to help create code, design, text, or other project material?

☒ Yes ☐ No

If yes, list them below:

Tool & Version	**Used For**	**Extent of Use (minor / moderate / major)**
Claude Code (Anthropic) [Claude Sonnet 5.5 and Opus 5.5]	Most of the backend, planning engine, tests and documentation, and drafting the submission text	Major
Replit Agent (Replit)	User-interface layout and styling, and fixes made by teammates inside Replit (about a third of the commits)	Moderate

Briefly describe how your team reviewed, tested, and took responsibility for AI-generated output:

[Maximum 100 words]

Changes were committed in small steps and covered by automated tests: 289 backend tests and 12 frontend tests. The forecast and buffer maths are checked against a separate reference prototype and a forecast audit on simulated restaurants. Commits made by the Replit Agent are fetched, built and checked before they are merged into the main branch. The team ran the app to check each screen, and edited AI-written document text. Every team member is responsible for being able to explain each component.

5. Sponsor-Provided Resources

Declare any official SDC sponsor-provided GenAI API, cloud credits, tools, mentoring, or data that your team used.

Did your project use any official SDC sponsor-provided resource?

☒ Yes ☐ No

If yes, list them below:

Sponsor	**Resource (API / credits / platform / data)**	**How It Was Used in the Project**
Replit	Replit credits	Used for development (workspace and Replit Agent) and for hosting the live prototype and its database

Was the sponsor resource essential to the core functionality of the prototype, or could it be replaced? Explain briefly:

[Maximum 100 words]

The Replit credits were not essential to the core functionality. Replit was used to build and host the prototype, and it is required by the event rules for deployment. The application itself is standard FastAPI, PostgreSQL and React, so it could run on another host by changing the database address and start command. The core journey (forecast, purchase plan, approval, supplier note) uses no AI and no sponsor service. The Replit Agent only sped up development; the team reviewed and tested what it wrote.

6. Security, Privacy & Responsible Use

6.1 Data Handling & Privacy

What user or personal data does your application collect, store, or send to third-party services (including AI APIs)? How is it protected, and how long is it kept? State "No user data is collected" if applicable.

[Maximum 150 words]

Collected and stored (PostgreSQL on Replit): fictitious demo users (name and role), sales, stock, recipes, suppliers with fake phone numbers, orders and an activity log of who did what. All of it is synthetic; no real personal data is used. Sent to third parties: Help chat questions go to OpenRouter and the model provider with the person's role and current page, but no business data. Uploaded PDF sales reports are also sent to OpenRouter for reading; CSV and Excel files are read inside the app and never leave it. API keys are kept in Replit Secrets, not in the code. Retention: demo data stays until the owner resets it; unsaved upload previews are deleted after one day; removed sales stay until the owner deletes them forever. A real deployment would follow Malaysia's PDPA: consent, limited purpose, retention limits and deletion on request.

6.2 Known Limitations & Risks

Identify the main technical limitations of your AI components (e.g., hallucination, bias, rate limits, dependency on a single provider) and any mitigation you have implemented.

[Maximum 150 words]

Hallucination and misreading: the language model can misread a PDF or answer a Help question wrongly. Mitigations: its answer must match a strict format; PDF rows are only a draft that a person checks and edits in a preview before anything is saved; dishes it cannot match are left out, never guessed; the Help chat answers common questions from a fixed list, otherwise only from our own help text, and cannot see or change business data. Rate limit: three typed Help questions per person per minute. Single provider: AI depends on OpenRouter and one configured model; if it fails, the app shows a plain message and manual forms plus CSV and Excel upload still work. The forecast uses no AI, but sold-out days or unannounced promotions can make it wrong, so every line shows its safety buffer and a person approves each order. Sign-in is simulated in this prototype.

7. Compliance & Readiness Checklist

Tick all boxes that apply before submitting.

- ☑ All APIs, cloud services, AI tools, models, and datasets used in the project are declared in this document.
- ☑ All third-party code, datasets, and models are used in accordance with their licenses, and attribution is provided where required.
- ☑ No API keys, passwords, or other credentials are exposed in the code repository, prototype, or this document.
- ☑ Any use of generative-AI tools for development has been disclosed and reviewed by the team.
- ☑ Any use of official SDC sponsor resources has been declared in Section 5.
- ☑ The prototype described here is accessible to judges through the links provided in Section 1.
- ☑ All fields in this template have been completed within the word limits stated.

8. Declaration

By submitting this document, the team leader confirms that the information provided is complete and accurate to the best of the team's knowledge, and that every API, cloud service, AI tool, dataset, model, and sponsor resource used in the project has been declared.

[Signature]



[Date]

25/10/26